



SpaceX's Starship

One of the raptor engines has been swapped out for the SpaceX Starship SN10. <http://digit.in/mar21-93>



Dark Matter

HAYSTAC, the dark matter detector, uses quantum technology to scan for dark matter particles. <http://digit.in/mar21-94>

invent the modern understanding of special relativity.

EINSTEIN AND HIS ANNUS MIRABILIS

In 1905, Einstein published four papers, which came to be known as the Annus Mirabilis papers. His first paper on the photoelectric effect worked on the particulate effect of light, and it was understood from the postulate that light interacts with matter in the form of discrete packets called quanta of energy. It was already an idea put forward by Max Planck in 1900 as a pure mathematical coincidence which contradicted the contemporary theory of light (as a wave). This is the only work Einstein considered to be revolutionary. His second paper on Brownian Motion explained the random movement of small particles with evidence of molecular action which further supported the atomic theory. His third paper on electrodynamics introduced the radical idea of special relativity which showed that the observed speed of light was dependent on the observer's state of motion itself. This implied that the time-space frame of a moving object slowed down and contracted relative to the observer's frame of reference. In his final paper, on mass-energy equivalence, Einstein went on to show from his equations of special relativity that $E=mc^2$ which suggested that small amounts of mass could be converted to massive amounts of energy. These papers are considered to be tremendous achievements in physics and hence is known as his Annus Mirabilis "Wonderful Year".

MID-20TH CENTURY

Paul Dirac, in 1920, first formulated the quantum theory which described the interaction of radiation and matter. He was able to compute the coefficient of spontaneous emission of an atom. He described the quantization of the electromagnetic field. Further research by Wolfgang Pauli, Eugene Wigner, Pascual Jordan, and of course, Enrico Fermi's beautiful formulation of quantum electrodynamics, the scientific world accepted and believed that in principle, it was pos-

sible to compute quantities of physical processes that involved photons and charged particles both. However studies in 1937 revealed that such computations were not reliable. Infinities emerged at higher order processes which made the computations meaningless and casted doubts on the fundamental compatibility between special relativity and quantum mechanics. These difficulties were only further exacerbated through the end of 1940, where the advances in microwave technology now made it possible to take precise measurements of the level shifts of a hydrogen atom, and the magnetic moment of an electric. These experiments showed discrepancies which the theory was unable to explain, and at the same time in the 1950s, bubble and spark chambers were invented to discover particles known as hadrons.

The solution to the electron problem of hydrogen was given by Hans Bethe in 1947, in which he completed the first non-relativistic computation of the line shifts of the hydrogen atom. Despite its limitations, it agreed very well with the theory. His idea was to attach infinities to the mass and charge that were actually fixed values, which caused the infinities to be absorbed in the constants, thereby giving finite results – this process was named renormalization. Based on Bethe's work, as well as the fundamental papers by Julian Schwinger, Shinichiro Tomonaga, Freeman Dyson and Richard Feynman, it was finally possible to get covariant formulations that were finite at any order in a perturbation series of QED. Feynman's mathematical technique in particular was groundbreaking. Based on his diagrams, initially it seemed quite different from the field-theoretic but Freeman Dyson later showed that both approaches were one and the same. Renormalization subsequently became one of the fundamental aspects of quantum field theory. QED has served as the model and template for all subsequent quantum field theories.

Peter Higgs, Jeffrey Goldstone and others independently showed how the weak nuclear force meets quantum electrodynamics in a single electroweak force. Sheldon Glasgow's discovery in

1960, of a way to combine the electromagnetic and weak interactions was the first step towards the development of the Standard Model. In 1967, Steven Weinberg among others, used the Higgs mechanism and incorporated it into the electroweak theory, which is the form in which we know it as today. The Higgs mechanism is believed to give mass to all elementary particles in the Standard Model, including the W and Z bosons, quarks and leptons. This model was confirmed by the neutral weak currents caused by a Z boson exchange, as discovered at CERN in 1973, which made the electroweak theory widely accepted. In 1981, the W and Z bosons were discovered experimentally and their masses were as predicted by the Standard Model. The strong interaction, responsible for the strong nuclear force, got its modern form in 1973 after hadrons were experimentally confirmed to be composed of fractionally charged quarks. Quantum Chromodynamics was established in the 1970s, and arrived at a set of fundamental particles that established the Standard Model – it described all forces except for gravity.

21ST CENTURY

Our modern use of electricity far exceeds what humans have ever used in history. The scope of emerging energy technologies has grown exponentially ever since the advent of the invention of the nanowire, a lithium-ion battery and electric double layer capacitors. We also have medical applications in the form of an MRI (Magnetic Resonance Imaging). Advances in wireless energy transfer has enabled the ability to provide electrical energy to objects without wires, this has applications in diverse fields such as consumer, commercial, medical and military. Finally, our thirst of arriving at one theory – a "Theory of Everything" continues. A Grand Unified Theory (GUT) is a model that attempts to combine the electromagnetic force, with the weak and strong nuclear forces. Although multiple candidates for such a theory have been suggested, none of them have been accepted by the physics community. 